Theoretical Skills

International Physics Olympiad · Worked Solutions

Introductory study guide · 14 original problems

> Organized around the IPhO syllabus. This guide is not exhaustive; consult the syllabus study PDF and the IPhO syllabus webpage for the full scope and qualifications.

These worked solutions accompany the original MAIS PAC problems. Check your assumptions and units.

1. A body starts from rest and accelerates uniformly at 2 m/s² for 3 s. Find its speed and displacement.

   Solution. $v = at = 6$ m/s; $x = at^2/2 = 9$ m.

2. A 2 kg mass moves at 3 m/s. Find its kinetic energy.

   Solution. $K = mv^2/2 = 9$ J.

3. A 12 V ideal battery is connected across a 4 Ω resistor. Find current and power.

   Solution. $I = V/R = 3$ A; $P = VI = 36$ W.

4. A wave travels at 12 m/s with frequency 3 Hz. Find its wavelength.

   Solution. $\lambda = v/f = 4$ m.

5. A thin converging lens has f = 10 cm and an object at u = 30 cm. Find the image distance.

   Solution. $1/v = 1/10 - 1/30 = 1/15$ cm^{-1}, so $v = 15$ cm, a real image.

6. A clock moves at 0.6c relative to a lab. How much lab time passes during 4 s of proper time?

   Solution. $\gamma = 1/\sqrt{(1-0.6^2)} = 1.25$, so $\Delta t = \gamma \Delta\tau = 5$ s.

7. A radioactive sample has a half-life of 2 hours. What fraction remains after 6 hours?

   Solution. Three half-lives pass, leaving $(1/2)^3 = 1/8$.

8. An ideal heat engine operates between 600 K and 300 K. What is its maximum efficiency?

   Solution. The Carnot limit is $1 - 300/600 = 0.50$, or 50%.

9. Challenge 1 — A small block is released from rest at height h on a frictionless fixed track. At the bottom it collides and sticks to a stationary block of equal mass on a horizontal frictionless surface. Find their common speed and the fraction of the original mechanical energy converted to internal energy in the collision.

   Solution. Before impact, energy gives $v = \sqrt{(2gh)}$. During the short collision, horizontal momentum is conserved: $mv = 2mV$, so $V = \sqrt{(2gh)}/2$. The final kinetic energy is $(1/2)(2m)V^2 = mgh/2$, half the initial mgh. Thus half is converted to internal energy. Kinetic energy is not conserved in the sticking collision.

10. Challenge 2 — A satellite is in a circular orbit of radius r around a spherical body of mass M. A brief tangential impulse reduces its speed to a fraction α of the original circular speed, where $0 < \alpha < 1$. Find the semimajor axis a of the new orbit and identify whether the impulse point is periapsis or apoapsis.

    Solution. Initially $v_0^2 = GM/r$. The specific energy after the impulse is $\varepsilon = \alpha^2 GM/(2r) - GM/r = -GM(2-\alpha^2)/(2r)$. Using $\varepsilon = -GM/(2a)$ gives $a = r/(2-\alpha^2)$. Radial speed is zero at the impulse and speed is below circular, so this point is apoapsis. The $\alpha \to 1$ limit gives $a \to r$; $\alpha = 0$ is excluded because the orbit degenerates into radial fall.

11. Challenge 3 — Two identical capacitors of capacitance C are isolated. One initially has voltage V; the other is uncharged. They are connected in parallel with matching polarities by idealized conducting wires. Find the final voltage and the energy dissipated during redistribution.

    Solution. Charge is conserved: initial charge CV becomes $2CVf$, so $Vf = V/2$. Initial energy is $CV^2/2$; final energy is $2 \times C(V/2)^2/2 = CV^2/4$. The difference $CV^2/4$ is dissipated or radiated during the transient. An exactly lossless, zero-inductance connection cannot describe this transient physically.

12. Challenge 4 — A source emits light of vacuum wavelength λ. In a double-slit experiment with slit separation d and screen distance $L \gg d$, determine the distance between neighbouring bright fringes near the centre. State the approximation.

    Solution. At angle θ the path difference is approximately $d \sin\theta$. Adjacent maxima differ by λ in path difference. For small θ, $\sin\theta \approx \tan\theta \approx y/L$, so $\Delta y \approx \lambda L/d$. This assumes coherent slits and paraxial geometry.

13. Challenge 5 — A particle at rest of rest mass M decays into two photons travelling in opposite directions. Find each photon’s energy. What changes if the photon energies are unequal in the laboratory?

Solution. In the rest frame initial momentum is zero. Opposite photon momenta must have equal magnitudes, so their energies are equal. Energy conservation gives $E_1 = E_2 = Mc^2/2$. In a laboratory frame where the parent moves, photon energies need not match; the total energy and momentum still transform and are conserved.

14. Challenge 6 — One mole of a monatomic ideal gas expands quasistatically and isothermally from V to 2V at temperature T. Find the work done by the gas and heat absorbed. Then compare its entropy change with that of a reservoir at the same T.

Solution. Since $p = RT/V'$, $W = \int_{v}^{2v} RT\, dV'/V' = RT \ln 2$. For an ideal gas internal energy depends only on temperature, so $\Delta U = 0$ and $Q = W$. Gas entropy rises by $Q_{rev}/T = R \ln 2$; a reservoir at the same temperature loses $R \ln 2$. In the reversible limit total entropy change is zero.